Throughout my undergraduate journey at Iowa State, I have worn many faces. I have been a TA for CPRE 2880: Embedded Systems, I have been in leadership in two engineering clubs and the honors program, I have pursued a computer engineering major and a mathematics major, I have attended almost every ISU theatre performance, and taken many courses offered by the theatre program. I have been a research aid, an embedded software intern, and a web developer.

When I first started college, I was wary about what I actually wanted to do with a computer engineering degree. I had some ethereal idea that I wanted to work on computer chips, but I didn't really know what that would mean for my future, or what classes would lead me to that goal. I worked with my advisor to plan out my first year of classes, in which I would learn C, expand on my Java knowledge base. Notably, I took MATH 201: Introduction to Proof and THTRE 255: Introduction to Theatrical Production. At its core, THTRE 255 is about the backstage of a theatre performance, from the set to the lighting, costumes to props, etc. This course let me explore the softer skills required of every person, such as interpersonal communication and problem solving. One of my favorite memories from this course was making a marbling texture on a sheet of wood by watering down paint and splattering it across the plane, then watching it dry into a convincing replica of stone.

MATH 201 changed by academic plan for the rest of my time at Iowa State. It formalized many of the structures that I have seen in mathematics throughout my life, like that a function isn't just a thing that takes input and give output; formally a function f is a subset of the domain $\times$ the codomain such that some element x in the domain only has at most 1 element in the codomain in the subset. Applying this level of thinking to writing and debugging software and HDL code in my courses allowed me to formalize my thinking, and overall reduce errors that I ran into. MATH 201 showed me that mathematics has much to offer me that I found interesting, such as Graph Theory and Abstract Algebra, which can be mapped to common problems in Computer Engineering, but it also taught me to enjoy proving something to prove it, not just to use it.

In the next semesters, I would continue down the Computer Engineering Curriculum, but outside of my classes I continued by work in the Cyclone Rocketry club, where I helped design and code flight control hardware to track a rocket's energetic ascent into the sky, and its descent back to the ground. This club let me apply what I learned in my courses in a different context, and conversely it taught me more about my classes before I took them. I strongly credit this club, and my involvement in it, with my success in becoming and staying an embedded systems TA.

In my third year here, I started my lab TA position for CPRE 288 in the beautiful Coover 2041. In this position, I got to help students learn about the consequences of writing sloppy code, and the consequences of letting AI write all of your code. From 2024 to now, I have worked with hundreds of students, and I have noticed that the average student relies on AI as a crutch more and more, and seems to actually understand the contents of the course less and less, and I predict that over the next years, we will see more fails and drops

of courses, as courses have to adjust their exams and syllabi to appropriately ensure students are meeting the learnign objectives. Seeing this decline in AI has fueled me to avoid it during the learning process to ensure that I actually understand the content as it is presented, and can use it in the future. Being a TA also let me interact with a diverse set of students. I have seen students who have been programming for a decade fly through labs, and I have seen electrical engineering majors do everything they can to avoid writing more code than they need to. I have helped students from around a dozen different nations understand the course content in personalized explanations.

In years 4 and 5, I have been in leadership for a new club on campus: Chip Forge. Last year, as vice-president, I merged my senior design project with the clubs structure, and we were awarded technical achievement by the department for our custom computer chip that we designed: μGPU. Through this club, I have further realized the true expenses required for ASIC fabrication. This year as president, we are running out of NSF grant money. Now, in order to maintain our status as a club that tapes out, we must reach out to companies to ask for funding for tapeout. While our senior design project may have been extremely successful, keeping the club active requires a whole new set of soft skills that my years as a TA and in club leadership have helped me with: working with and managing other people.

Overall, I have enjoyed my time at Iowa State, and the experiences that I have had. All of the faces that I have worn cumulate into who I am today, and I am grateful for all of them and how they have affected me. I am leaving this degree as a very different person than who started it, and I feel that at the end of this degree I will be fit to call myself a computer engineer.